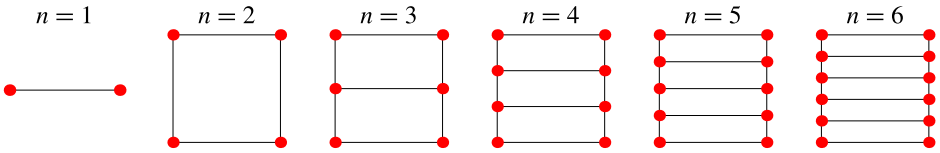
Ladder Graph

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The n-ladder graph can be defined as $L_n = P_2 \square P_n$, where P_n is a path graph (Hosoya and Harary 1993; Noy and Ribó 2004, Fig. 1). It is therefore equivalent to the $2 \times n$ grid graph. The ladder graph is named for its resemblance to a ladder consisting of two rails and n rungs between them (though starting immediately at the bottom and finishing at the top with no offset).

Hosoya and Harary (1993) also use the term "ladder graph" for the graph Cartesian product $K_2 \square C_n$, where K_2 is the complete graph on two nodes and C_n is the cycle graph on n nodes. This class of graph is however more commonly known as a prism graph.

Ball and Coxeter (1987, pp. 277-278) use the term "ladder graph" to refer to the graph known in this work as the ladder rung graph.

The ladder graph $P_2 \square P_n$ is graceful (Maheo 1980).

The chromatic polynomial, independence polynomial, and reliability polynomial of the ladder graph L_n are given by

$$\pi_n(z) = (z-1)\, z \left(z^2 - 3z + 3\right)^{n-1} \tag{1}$$

$$I_n(z) = 2^{-(n+1)} \left[(s - 3x - 1)(x - s + 1)^n + (s + 3x + 1)(x + s + 1)^n\right] \tag{2}$$

$$C(p) = \frac{(p-1)^{2n-1}}{2^n \sqrt{p(9p+2)+1}} \left[\left(3p - \sqrt{p(9p+2)+1} + 1\right)^n - \left(3p + \sqrt{p(9p+2)+1} + 1\right)^n\right], \tag{3}$$

where $s = \sqrt{x^2 + 6x + 1}$. Recurrence equations for the chromatic polynomial, independence polynomial, matching polynomial, rank polynomial, and reliability polynomial are given by

$$\pi_n(z) = \left(z^2 - 3z + 3\right) \pi_{n-1}(z) \tag{4}$$

$$I_n(x) = (x+1)\, I_{n-1}(x) + x\, I_{n-2}(x) \tag{5}$$

$$\mu_n(x) = \left(-2 + x^2\right) \mu_{n-1}(x) - x^2\, \mu_{n-2}(x) + \mu_{n-3}(x) \tag{6}$$

$$C_n(x)=(3x+1)C_{n-1}(x)-xC_{n-2}(x). \quad (8)$$

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REFERENCES

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